EXCELLIGENT

SAP Middleware Integration Requirements

Introduction

The objective of the SAP Middleware is to securely and efficiently synchronize data between the SAP system and a SQL database. The middleware acts as a bridge between SAP and downstream applications by automatically fetching data from SAP, processing it, and storing it in SQL. This reduces direct dependency on SAP, improves application performance, and provides centralized monitoring and management of data synchronization.

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1. Middleware Responsibilities (Our Scope)

1. React Admin UI

A web-based dashboard will be developed to manage the integration. It will allow administrators to:

- Configure SAP and SQL connections
- Select SAP tables for synchronization
- Map SAP fields to SQL fields
- Configure synchronization schedules
- View synchronization logs and job status
- Manage users and access permissions

2. Node.js/Express Middleware

The backend middleware will connect SAP with the SQL database. Its responsibilities include:

- Reading data from SAP
- Processing and validating data
- Applying business rules
- Synchronizing data with SQL
- Handling errors and retries
- Maintaining synchronization logs

3. SQL Database

A SQL database will store the replicated SAP data. This improves application performance, reduces load on SAP, and enables faster reporting and integration with other systems.

4. Synchronization Engine

The middleware will support two synchronization modes:

- **Initial Sync:** Transfers all existing records during the first execution.
- **Incremental Sync:** Transfers only newly added or modified records during subsequent executions.

5. Scheduler

The scheduler will automate synchronization based on business requirements. Supported options include:

- Hourly
- Daily
- Weekly
- Custom schedule (Cron Expression)

6. Duplicate Detection

The middleware will use the table's primary key to identify duplicate records.

- If the record exists → Update
- If the record does not exist → Insert

This ensures data consistency.

7. Insert, Update & Delete Rules

The middleware will support:

- Insert new records
- Update existing records
- Handle deleted records based on the agreed strategy (Delete, Soft Delete, or Ignore)

8. Batch Processing

Large volumes of data will be processed in batches instead of one record at a time. This improves performance, reduces memory usage, and makes synchronization more reliable.

9. Logging & Monitoring

Every synchronization job will be logged with details such as:

- Start and end time
- Table name
- Records processed
- Inserted, Updated, Failed records
- Job status
- Error details

These logs help administrators monitor system health and troubleshoot issues.

10. Retry Mechanism

If synchronization fails due to temporary issues such as network interruptions, the middleware will automatically retry the failed operation based on configurable retry settings.

11. User Authentication & Roles

The application will implement Role-Based Access Control (RBAC).

Typical roles include:

- **Admin:** Full configuration access
- **Operator:** Execute and monitor synchronization
- **Viewer:** Read-only access to dashboards and logs

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2. Information Required from SAP Team

To successfully implement the integration, the following information is required from the SAP team.

SAP Connectivity

- SAP Database, OData, API, RFC, or BAPI access

- Read-only user credentials
- Required network, VPN, or firewall access

Table Information

- List of SAP tables to be synchronized
- Primary keys for each table
- Relationships between tables
- Business rules and dependencies

Synchronization Details

- Last Modified/Timestamp field (if available)
- Expected synchronization frequency
- Test environment
- Sample data for validation

Technical Support

A technical contact from the SAP team should be available to assist with connectivity, authorization, and data-related issues during development and deployment.

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3. Joint Decisions

The following decisions should be finalized jointly by the Middleware and SAP teams.

Delete Strategy

Determine how deleted SAP records should be handled:

- Physical Delete
- Soft Delete (Inactive)
- Ignore Deleted Records

Conflict Handling

Define how conflicting updates should be resolved if the same data is modified in multiple systems.

Synchronization Schedule

Agree on synchronization timings to minimize system impact while meeting business requirements.

Error Notification

Decide how synchronization failures should be reported, such as:

- Email notifications
- Dashboard alerts
- Log monitoring

Security

Security measures should include:

- Secure communication (HTTPS/TLS)
- Encrypted credentials
- Role-Based Access Control
- Audit logging

Performance Expectations

Define expected performance, including:

- Number of records to be synchronized
- Maximum synchronization duration
- Batch size
- Acceptable response time

Conclusion

The SAP Middleware will provide a secure, scalable, and automated solution for synchronizing SAP data with a SQL database. It will include features such as configurable scheduling, incremental synchronization, duplicate detection, logging, monitoring, retry mechanisms, and role-based security. Successful implementation will require close collaboration between the Middleware and SAP teams to finalize connectivity, business rules, synchronization strategy, and security requirements.